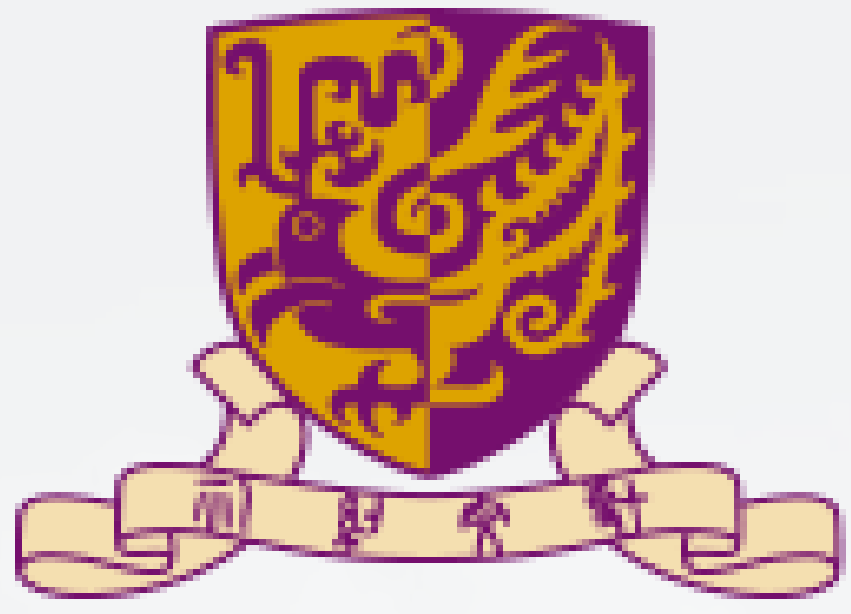




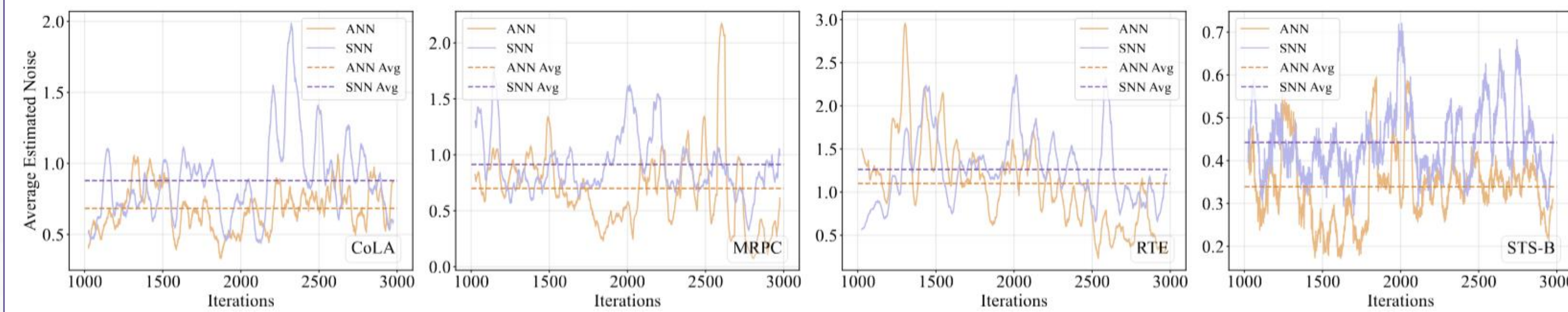
AdaS: Adaptive Gradient Descent for Spiking Transformers

Zijian Zhou, Honglin Cao, Ammar Belatreche, Wenjie Wei, Yimeng Shan, Yu Liang,
Yu Yang, Shuai Wang, Yalan Ye, Malu Zhang, Yang Yang, Haizhou Li



ICML
International Conference
On Machine Learning

Motivation



- Spiking Transformers are difficult to optimize because the non-differentiable spike function prevents direct gradient-based training.
- To quantify the resulting optimization distortion, we first define the ideal next-step parameter from a proximal perspective:

$$\theta_t^* = \arg \min_{\theta} \frac{\|\theta - \theta_{t-1}\|_2^2}{2\eta_t} + f(\theta).$$

- Given the actual update $r_t = \theta_t - \theta_{t-1}$, the parameter-update noise is defined as the extra proximal objective value induced by using r_t :

$$\mathcal{N}(r_t) = \frac{\|r_t\|_2^2}{2\eta_t} + f(\theta_{t-1} + r_t) - \frac{\|\theta_t^* - \theta_{t-1}\|_2^2}{2\eta_t} - f(\theta_t^*).$$

- Since θ_t^* is intractable, we use a computable mini-batch estimate:

$$\mathcal{N}_{\text{est},t} = \frac{\|r_t\|_2^2}{2\eta_t} + f_t(\theta_t) - f_t(\theta_{t-1}), f_t(\theta) = \frac{1}{|\mathcal{B}_t|} \sum_{(x,y) \in \mathcal{B}_t} \ell(\theta; x, y).$$

- Under a first-order approximation, the local noise-minimizing update is the negative-gradient step: $r_t^* = -\eta_t \nabla f_t(\theta_{t-1})$. Thus, parameter-update noise measures how much the actual optimizer step deviates from the locally optimal descent direction.
- For ANN Transformers, the mini-batch objective f_t is differentiable and can be directly optimized by backpropagation. In contrast, Spiking Transformers optimize a surrogate objective $\tilde{f}_t$, whose local noise-minimizing update is $\tilde{r}_t^* = -\eta_t \nabla \tilde{f}_t(\theta_{t-1})$, while the ideal descent update of the original spiking objective can be denoted as $r_{t,\text{spk}}^* = -\eta_t g_t^{\text{spk}}$. Therefore, the deviation of the actual SNN update from the ideal spiking-objective update can be decomposed into two parts:

$$r_t - r_{t,\text{spk}}^* = \underbrace{(r_t - r_t^*)}_{\mathbf{n}_{\text{opt}}} + \underbrace{(r_t^* - r_{t,\text{spk}}^*)}_{\mathbf{n}_{\text{sur}}}.$$

- Where $\mathbf{n}_{\text{opt}}$ is the optimizer-induced deviation and $\mathbf{n}_{\text{sur}}$ is the surrogate-induced discrepancy. The measured surrogate-level noise mainly reflects $\mathbf{n}_{\text{opt}}$. When these two components are weakly correlated, the total deviation approximately satisfies

$$\|r_t - r_{t,\text{spk}}^*\|_2^2 \approx \|\mathbf{n}_{\text{opt}}\|_2^2 + \|\mathbf{n}_{\text{sur}}\|_2^2.$$

- As shown in the Figure above, since the observable noise of Spiking Transformers is already higher than that of ANNs, and the additional $\|\mathbf{n}_{\text{sur}}\|_2^2$ further enlarges the total deviation, Spiking Transformers suffer from **excessive parameter-update noise**.

Method

- To mitigate excessive optimizer-induced deviation, AdaS combines the adaptive update direction u_t from the base optimizer with the first-order momentum m_t , leading to

$$\theta_t = \theta_{t-1} - \eta_t [\alpha_t u_t + (1 - \alpha_t) m_t].$$

- Using a first-order approximation of the estimated update noise and treating m_t as a stable approximation of the gradient direction, AdaS obtains

$$\mathcal{N}_{\text{est}}(\theta_t) \approx \frac{\eta_t}{2} (\alpha_t^2 \|u_t - m_t\|_2^2 - \|m_t\|_2^2).$$

- Instead of removing all update noise, AdaS sets a target noise level γ and solves for the adaptive balancing coefficient:

$$\alpha_t^* = \Pi_{[0,1]} \left(\frac{\sqrt{\|m_t\|_2^2 + 2\gamma/\eta_t}}{\|u_t - m_t\|_2} \right).$$

- This adaptive balancing suppresses excessive noise while preserving moderate exploration, and it can be integrated into existing adaptive optimizers without additional memory overhead by reusing their stored first-order momentum.

Algorithm 1 AdaS

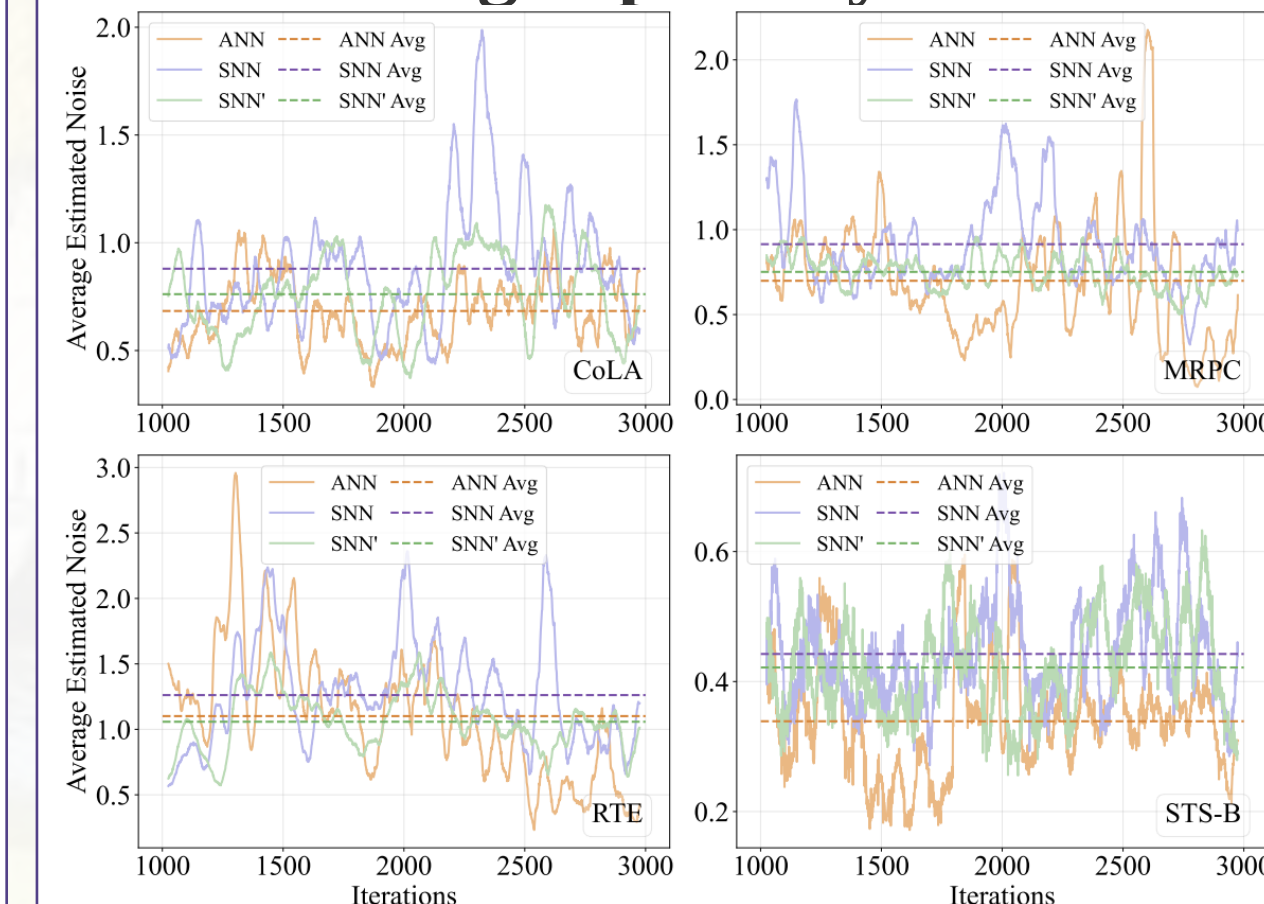
Require: Initial parameters θ_0 , learning rate $\{\eta_t\}_{t=1}^T$, weight decay factors $\{\lambda_t\}_{t=1}^T$, target noise level γ
Ensure: Optimized parameters θ_T
1: **for** $t = 1, \dots, T$ **do**
2: Compute gradient g_t
3: Update the first-order moment m_t
4: Compute adaptive update component u_t
5: Compute balancing weight: $\alpha_t \leftarrow \frac{\sqrt{\|m_t\|_2^2 + 2\gamma/\eta_t}}{\|u_t - m_t\|_2 + \epsilon}$
6: Project α_t onto $[0, 1]$: $\alpha_t \leftarrow \max(0, \min(1, \alpha_t))$
7: $\theta_t \leftarrow \theta_{t-1} - \eta_t [\alpha_t u_t + (1 - \alpha_t) m_t + \lambda_t \theta_{t-1}]$
8: **end for**

Experimental Results (Continued)

➤ ADE20K and CIFAR10-DVS

Dataset	Type	Method	Architecture	Param(M)	Time	Optimizer	MIoU/Acc.(%)
ADE20K	ANN	(Yu et al., 2022)	ResNet-18	15.5	—	SGD	32.9
		(Wang et al., 2021)	PVT-Small	28.2	—	AdamW	39.8
		(Wang et al., 2023a)	InterImage-T	59.0	—	AdamW	48.1
	SNN	(Yao et al., 2025)	E-SpikeFormer	20.4	4	SGD	4.6 [†]
				20.4	4	Lion	38.5 [†]
				20.4	4	AdamW	38.2 [†]
CIFAR10-DVS	SNN	(Zhou et al., 2026)	Spikingformer-2-256	2.6	16	AdamW	81.3
				2.6	16	AdamW	80.9
				2.6	16	AdamW	80.0
		(Zhou et al., 2023)	S-Transformer-2-256	1.5	16	SGD	20.9 [†]
				1.5	16	Lion	82.2 [†]
				1.5	16	AdamW	84.0
	SNN	(Zhou et al., 2024)	QKFormer-2-256	1.5	16	AdamW	84.0
				1.5	16	AdaS	85.1

➤ Denoising capability



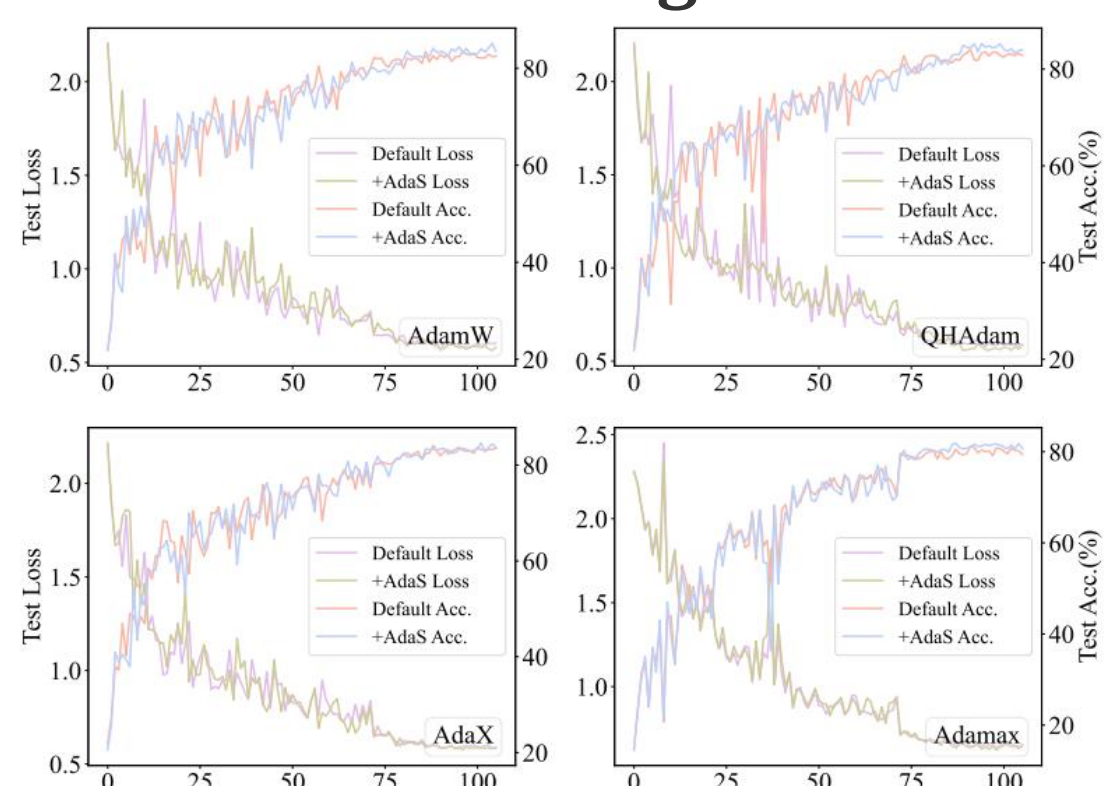
➤ Adaptive balancing

Optimizer	α	Performance(%)	
		CoLA	CIFAR10-DVS
AdamW	—	38.8	83.5 [†]
AdaS*	0.9	40.9(+2.1)	83.4(+0.1)
	0.95	38.9(+0.1)	84.2(+0.7)
	0.99	39.8(+1.0)	84.0(+0.5)
AdaS	Adaptive	42.1(+3.3)	85.1(+1.6)

➤ Versatility of AdaS

Optimizer	Setting	Performance(%)	
		CoLA	CIFAR10-DVS
AdamW	Default	38.8	83.5
	+AdaS	42.1(+3.3)	85.1(+1.6)
AdaX	Default	38.2	84.1
	+AdaS	40.5(+2.3)	84.6(+0.5)
QHAdam	Default	38.9	84.0
	+AdaS	40.4(+1.5)	85.3(+1.3)
Adamax	Default	39.2	81.2
	+AdaS	40.1(+0.9)	81.9(+0.7)

➤ Rate of convergence



Experimental Results

➤ GLUE benchmark

Method	Optimizer	MNLI _{m/mm}	QQP _{F1}	QNLI	SST-2	CoLA	STS-B	MRPC _{F1}	RTE	Avg.
(Zhang et al., 2020)	AdamW	47.2/47.3	67.0	61.3	80.6	0.0	4.7	81.2	52.7	49.1
(Devlin et al., 2019)	—	83.8/83.4	90.5	90.7	92.3	60.0	89.4	89.8	69.3	83.2
(Xing et al., 2024)	AdamW	56.8/55.2	70.0	60.6	80.6	14.6	20.0	82.3	53.8	54.9
(Lv et al., 2025)	AdamW	71.4/71.0	68.2	66.4	85.4	16.9	18.7	82.0	57.5	59.7
(Zhou et al., 2026)	AdamW	70.2/70.6	80.9	79.5	83.9	12.8	77.0	83.0	62.1	68.9
(Xing et al., 2024)	Lion	76.6/77.0	84.4	84.3	86.0	37.0	85.4	84.8	69.3	76.1
	AdamW	77.1/77.2	83.9	85.3	87.0	38.8	84.9	85.7	69.0	76.5
	AdaS	77.0/77.6	84.8	85.7	88.0	42.1	85.6	87.0	70.4	77.6

➤ Event-based tracking benchmarks

Methods	Optimizer	Param. (M)	Spiking Neuron	Timesteps ($T \times D$)	FE108		VisEvent	
					AUC(%)	PR(%)	AUC(%)	PR(%)
(Yan et al., 2021)	-	28.23	-	1 × 1	57.4	89.2	34.1	46.8
(Wei et al., 2023)	-	202.56	-	1 × 1	56.6	88.5	33.0	43.8
(Kang et al., 2023)	-	42.22	-	1 × 1	55.9	88.5	34.6	47.6
(Zhang et al., 2022)	-	20.55	LIF	3 × 1	-	-	35.0	50.3
(Zhang et al., 2025a)	-	31.40	BA-LIF	5 × 1	-	-	35.4	50.4
(Shan et al., 2025)	AdamW	19.61	I-LIF	1 × 4	59.0	91.3	35.6	49.2
	AdaS	19.61	I-LIF	1 × 4	60.2	92.5	36.3	50.5

Conclusion

- This work identifies excessive parameter-update noise as a key optimization bottleneck in training Spiking Transformers.
- Through a quantitative definition of parameter-update noise, the paper shows that Spiking Transformers exhibit stronger update deviation than ANN Transformers under comparable training settings.
- To address this issue, AdaS adaptively balances adaptive optimizer updates with a momentum-based gradient component, maintaining the update noise at an appropriate level.
- Extensive experiments on both computer vision and natural language processing tasks demonstrate that AdaS consistently improves the performance of various Spiking Transformers.